

Emojis vs. Text: Unpacking Emotional Conflicts in Modern Digital Communication Forums

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Abstract—This study investigates how well young adults interpret emotional sentiments in digital messages when the textual and visual cues are incongruent. Fifty participants completed an emotional classification survey with texts from three conditions: congruent (text and emoji sentiments matched), contradictory (text and emoji sentiments conflicted), and control (text only). The primary sample consisted of college-aged individuals between 18 and 24 years old, with an additional group of participants over the age of 24 included as a comparison. Participants judged each message as positive or negative, rated their confidence on a five-point Likert scale, and had their reaction times recorded. Accuracy was substantially higher in the congruent condition ($M = 94.6\%$) than in the contradictory condition ($M = 68.6\%$), and reaction times were longer in contradictory trials. In the contradictory condition, 31.4% of responses were misclassified, with nearly each error reflecting the emoji’s valence rather than the text’s, providing evidence of a dominance towards visual elements during conflict resolution. Individual question error rates ranged from 0% to 66%, suggesting that the visual dominance is calibrated by the relative strength of the competing signals at play. These findings contribute to a theoretical understanding of multimodal conflict resolution and carry practical implications for both natural language processing and future social media platform design.

Index Terms—emojis, multimodal learning, emotional analysis, cognitive science, conflict resolution

I. INTRODUCTION

Digital communication has undergone a drastic and fundamental transformation over the past two decades. Text messaging, once limited to purely digital language exchanged between individuals, has evolved into a rich communication medium in which words and images coexist and complement one another within a single message. At the center of this evolution sits the emoji, an expressive and visual icon that users can use to signal tone, modify meaning, convey humor, and express emotion to a degree that words alone cannot match. Research shows that emojis serve as a primary mechanism for emotional expression in online settings, functioning in ways that can mimic facial expressions and other visual cues in normal face-to-face interaction [4]. Among young adults in particular, emojis have become so deeply embedded in everyday communication that sometimes even the absence of them can carry a meaning in and of itself. A message sent without an emoji in a context where one would normally appear may be read as cold or passive-aggressive even when that was not the original intent of the message [9]. This shift in communication represents

not only a stylistic change in how people communicate with one another but a cognitive one as well. In today’s age of texting, recipients of digital messages must now integrate two distinct streams of information, verbal and visual, into a single interpretation of what the sender meant and felt.

In most cases this integration proceeds smoothly and without a lot of conscious effort. For example, a text message that says “I’m so happy for you! :)” would not present a challenge to interpret to nearly anyone. The text and the emoji communicate the same emotional valence, the cognitive system would then receive these reinforcing signals from the two channels, and the recipient is able to arrive at a confident judgment with a fairly minimal processing cost. Research in dual coding theory [3] predicts exactly this, that when verbal and visual representations are congruent, they reinforce each other and speed up efficient processing. However the modern emoji has also enabled a more complicated communicative phenomenon among young adults, that which is the deliberate or accidental misalignment of text and visual cues. Another example could be a text message that says “Oh great, another Monday :D” as it asks the recipient to reconcile a sarcastic or textual sentiment with a laughing, positive emoji. Whether the sender intended irony, sarcasm, or humor is unclear at a glance, and the recipient must now undergo a cognitive arbitration process before it can arrive at an interpretation for the message. Furthermore, research on sarcasm in digital communication shows that emojis are frequently deployed as a sign of ironic intent [5], which means that the interpretation of emoji and text mismatches is a common communicative challenge.

Understanding how individuals navigate that arbitration is important for several reasons. From a cognitive science perspective, emoji and text conflict provides a view at the mechanisms of multimodal integration and conflict resolution. The emoji and text pairing is an ideal environment for studying these processes as the two channels are easily separable, the stimuli are easily constructed, and the responses of interest (accuracy, speed, and confidence) are easily measurable.

From a practical perspective, this question also carries direct implications for the design of future social media and communication platforms, as well as the construction of emotionally intelligent AI systems. As machine learning models increasingly evolve and try to infer emotional intent from digital texts, the question of how humans on their own weigh these

competing signals will become a key concern [7]. A model that treats text and emoji as separate but equally weighted features may perform well when the two are congruent but that same model could systematically fail when they conflict.

This project addresses the following research question: How accurately do young adults judge emotional sentiment in text messages when emojis convey contradictory emotional information, compared to congruent emoji and text digital messages? As a member of the first generation to grow up with smartphones as a primary mechanism of social interaction, I am personally motivated by the frequency with which emoji and text mismatches have produced misunderstandings in my own experience. A laughing emoji appended to a complaint, a sad face following good news, a neutral face at the end of an enthusiastic message, etc. These combinations have at one point or another created a level of confusion that a purely verbal exchange would have avoided. Prior research supports the intuition that these mismatches matter as Riordan [1] demonstrated that emojis function as strong emotional modifiers in digital communication, and are capable of dramatically affecting the perceived tone of an accompanying message even when the textual content is clear. This study builds directly on that foundation by examining the cognitive costs of when those modifiers contradict the textual valence and force a breakdown in integration.

II. COGNITIVE SCIENCE FRAMING

The key foundation of this study draws on two connected bodies of work in cognitive science, dual coding theory, which explains how verbal and visual information are separately processed and integrated, and research on visual salience, which explains why visual stimuli may command disproportionate cognitive resources under competitive conditions. Together these frameworks generate specific and testable predictions about how the mind should respond when visual and verbal emotional signals conflict, and provide a basis for interpreting the patterns that emerge from the data.

A. Dual Coding Theory

Dual coding theory, which was originally proposed by Allan Paivio and elaborated by Clark and Paivio [3], holds that human cognition operates through two separate but still connected systems. They described how the verbal system processes *logogens*: words, sentences, and their relationships, and how the non-verbal system processes *imagens* (mental images): symbols, icons, etc. These systems operate in parallel and often results in an additive effect as it enables more robust processing than an encoding through only one of these channels.

For the purposes of this study, dual coding theory provides an insight into what happens when someone reads a digital message containing both text and an emoji. The verbal system would engage with the textual content of the message and produce a representation of it's emotional valence. Simultaneously, the non-verbal system would process the emoji as a symbol and produces its own emotional valence. Under

congruent conditions, both of these systems would output the same emotional valence and the integration would be straightforward as the two channels reinforce each other and provide the simple conclusion. Under contradictory conditions however it is not as simple as the two systems output opposing valences. The cognitive system must then engage in an emotional arbitration between the conflicting logogen and imagen. This additional computation is predicted to create a cognitive bottleneck, which will consume additional resources and time, which will be observable through a higher reaction time compared to the congruent conditions.

B. Visual Salience Research

The second pillar of this study comes from research on visual attention and its susceptibility to stimulus. Theeuwes [2] conducted research which demonstrated that visually salient stimuli or those that are colorful, or unexpected relative to their context, can capture resources in a manner that operates independently of the observer's goal. Importantly, this capture can also persist even after extensive practice and training, which suggests that the salience of the stimulus drives the allocation of attentional resources, rather than its actual relevance to the task.

Emojis are strong candidates for this kind of capture within communication environments. A text message is typically composed of alphanumeric characters, displayed in a single font and size. An emoji embedded within that message is a colorful and complex symbol that differs perceptually from its surroundings. If we use the logic of Theeuwes's account [2], we can see that the emoji functions as a distractor that captures attentional resources in a *bottom-up* fashion. Because the emoji contains intense visual features that differ from the standard alphanumerics, it triggers a shift of attention. This suggests that the emoji may dominate the final judgment not because it is more informative, but because its *perceptual salience* overrides the signals of relevance to the task.

This combination of dual coding theory and visual salience research helps to generate a prediction about the direction of errors in the contradictory condition. If participants are unable to resolve the conflict between the two channels, their errors should not be randomly distributed. They should be systematically biased toward the emoji, because attentional capture dramatically amplifies the influence of the visual cue during the arbitration process.

III. EXPERIMENT DESIGN

A. Participants

Fifty participants were recruited for this study. The primary sample consisted of 40 college-aged individuals between 18 and 24 years old, representing approximately 80% of the total sample. An additional 10 participants over the age of 24 were included as a secondary group, representing approximately 20% of the sample, in order to provide a comparison across a broader age range. Participants were recruited through various communication threads at the University of Virginia and through personal networks. Participants were given a brief

questionnaire prior to their enrollment to gauge their usage of emojis across various messaging platforms. This criteria was also established to further analyze whether participants' prior usage and interactions with emojis played a role in their ability to accurately classify the messages. The prediction was that participants who use or encounter fewer emojis in digital messages would struggle more than those who use them on a daily basis.

Participants were sometimes verbally sometimes textually informed of the nature of the task prior to beginning the experiment, including that their response times, classifications, and confidence ratings would be recorded for academic analysis. Participation was voluntary as noted on the site, and participants were free to discontinue at any time. Basic demographic information including age and emoji use frequency was collected from the initial questionnaire. No sensitive personally identifiable information or any institutional affiliations were collected at any stage. All response data was stored securely on an isolated machine and used exclusively for academic analysis.

B. Stimuli Creation

A set of 30 short text messages was constructed for the experiment, divided equally across three conditions of 10 stimuli each. Each stimulus was written to express a clear and unambiguous emotional valence either positive or negative, without any reliance on an emoji or visual element. Positive stimuli expressed enthusiasm and excitement. Negative stimuli expressed frustration and sadness.

In the *congruent* condition, a single emoji was appended to each message with the emotional valence of the emoji matching that of the text (EX., "I just got a promotion at work! :)") or "I failed the exam again :("). In the *contradictory* condition, an emoji of the opposing valence was appended (e.g., "That completely ruined my evening :D" or "I can't believe how well that went :("). In the *control* condition, no emoji was present. Emoji selection was limited to facial expression emojis with widely agreeable positive or negative valences in order to minimize ambiguity in the visual signals themselves [8].

Prior to using these emojis in the experiment, all 30 stimuli were reviewed by three independent raters who did not end up being participants in the main study. These raters assessed the intended emotional valence of each message from the text alone, and separately assessed whether the appended emojis in the congruent and contradictory conditions were perceived as intended. Stimuli that produced medium to high disagreement on in either condition were revised or replaced. The final set achieved full agreement on textual valence and on the alignment or misalignment of the emoji in its condition. Stimuli were then randomized in order separately for each participant to control any potential fatigue effects.

C. Procedure Details

The experiment was delivered through a custom web application that was designed specifically for this study and hosted

on a GitHub page accessible from any web browser. After completing a brief questionnaire including questions about age, cultural background, and self-reported frequency of emoji use across different platforms, participants navigate to the main experimental task through a clearly marked interface.

Each trial followed a rigid structure as a stimulus message was displayed in a format intended to resemble the visual appearance of a typical smartphone text message minus the quotation bubbles. This choice was deliberate as by presenting stimuli in a familiar format, the goal was to engage participants' natural thoughts rather than leading them towards a more analytical reading strategy. Participants were then instructed to classify the emotional tone of the message as positive or negative by clicking one of two labeled buttons positioned below the message bubble, and to do so as quickly and accurately as possible. Immediately following their classification response, participants selected their confidence level on a five-point Likert scale with 1 representing not confident and 5 being confident. Reaction time was recorded as the interval in seconds between the beginning of the stimulus message appearing and the moment the participant clicked their classification response.

The full scope of the experiment, including the questionnaire and 30 main trials, took approximately 10 to 15 minutes to complete on average per participant. All response data was logged automatically by the web application and exported in JSON format for statistical analysis.

D. Independent and Dependent Variables

The primary independent variable was the type of emotional alignment between the emojis and text, which was tweaked within subjects across the congruent, contradictory, and control levels. A secondary independent variable was the reported emoji use frequency, which was evaluated between subjects to examine individual differences. Age group (18 to 24 vs. over 24) was also treated as an additional factor of interest given the inclusion of participants outside the primary target range. The three dependent variables were classification accuracy, mean reaction time in seconds, and mean confidence rating on the Likert scale.

IV. RESULTS

TABLE I
PERFORMANCE METRICS ACROSS CONDITIONS

Condition	Accuracy (M)	Reaction Time (M)	Confidence (M)
Congruent	94.6%	7.15s	4.61
Control	—	9.15s	4.59
Contradictory	68.6%	9.47s	4.00

*Accuracy was not evaluated for the control condition, which served exclusively as a baseline for reaction time and confidence ratings.

A. Classification Accuracy

Classification accuracy was highest in the congruent condition ($M = 94.6\%$), and lowest in the contradictory condition

($M = 68.6\%$). This pattern deserves a second look of attention for two reasons. First the magnitude of the accuracy drop from congruent to contradictory (26 percentage points) is substantial, which suggests that conflicting emojis do impose a meaningful cost on emotional judgment beyond randomness. Secondly the individual variation visible in the contradictory condition indicates that some participants maintained relatively high accuracy, which presumably was by suppressing the visual signal and attending primarily to a holistic combined view, while still others showed severe accuracy declines consistent with unmitigated visual bias. Individual contradictory accuracy ranged from 30% to 100% across participants, a spread that is theoretically informative and explored further in the individual differences analysis below.

B. Reaction Time

Reaction times followed a pattern consistent with the dual coding prediction of processing costs under conflict. Contradictory trials produced the longest mean response times ($M = 9.47s$), exceeding both the control condition ($M = 9.15s$) and the congruent condition ($M = 7.15s$). The 2.32 second reaction time delay observed in contradictory trials provides evidence of the *processing* cost predicted by Clark and Paivio. When the logogen (text) and imagen (emoji) are incongruent, the brain cannot integrate them together and it must engage in a time-intensive conflict arbitration process to resolve the valence mismatch. This slowing down relative to what was observed in congruent trials is consistent with the prediction that deciding between two conflicting emotional valences consumes additional processing resources.

A paired samples t-test was also conducted to compare reaction times in congruent and contradictory conditions. There was not a significant difference in the scores for congruent ($M = 9.37$) and contradictory ($M = 7.42$) conditions; $t(49) = 1.70$, $p = 0.095$. These results suggest that in this sample while observable trends were seen, contradictory emojis did not impose a statistically significant cognitive delay compared to congruent pairings.

C. Confidence Ratings

Confidence ratings were highest in the congruent condition ($M = 4.61$ out of 5), followed closely by the control condition ($M = 4.59$), and lowest in the contradictory condition ($M = 4.00$). The drop observed in the contradictory condition is notable in that it occurred alongside the sharpest decline in accuracy and paired with the longest reaction times, which suggests that the conflict between channels did not go unnoticed by participants. Those who made errors in the contradictory condition were not doing so with high confidence, which indicates that the cognitive conflict imposed by the incongruent pairings rose to the level of conscious ambiguity.

D. Direction of Misclassifications

An analysis of errors in the contradictory condition revealed that 157 of 500 responses (31.4%) were misclassified. Given

the binary nature of the classification task, participants selecting the incorrect response often chose the option consistent with the emoji's emotional valence rather than the text's intended meaning. This pattern provides evidence that the visual cue was capable of overriding an unambiguous textual signal in over a third of contradictory trials, which supports the visual salience hypothesis that when the arbitration process fails to produce two consistent responses, the emoji dominates.

E. Individual Question Patterns

Considerable variation emerged at the individual question level within the contradictory condition. The three highest-error items were "I just found out I'm the only one in my friend group who got into the program" paired with a grimacing face emoji (66% error rate), "My thesis defense got moved up because my committee thinks it's ready" paired with an anxious face emoji (64.6% error rate), and "I cried less today than yesterday so I think I'm getting over it!" paired with a crying face emoji (60.4% error rate). In each case the emoji's emotional signal appears to have been crucial in overriding the positive textual content. By contrast, "My advisor hasn't responded to my emails in three weeks" paired with a relieved face emoji produced only two errors across all of the participants, which suggests that when the textual meaning is sufficiently strong, participants can still resist the visual signal entirely.

F. Individual Participant Differences

Self reported emoji use frequency showed no meaningful relationship with accuracy in the contradictory condition ($r = 0.004$, $p = .9758$). While it was hypothesized that heavier emoji users might develop stronger associations and in turn make it harder to suppress the visual signal, the data suggests that the visual dominance is an effect that operates independently of communicative habits. Similarly, no relationship was observed between emoji use frequency and performance in the congruent or control conditions. Another aspect came from the inclusion of participants over the age of 24 as notably, older participants with lower emoji use frequency, such as Individual One (age 65, Rarely) and Individual Two (age 62, Always), achieved high contradictory accuracy (90% and 80% respectively). This suggests that while familiarity does not dictate performance, age related strategies may warrant further investigation.

G. Control Accuracy Disclaimer

As shown in Table I, accuracy was not evaluated for the control condition. This was a deliberate decision to establish a clean baseline for reaction time and confidence. By treating the control condition as a reference point for cognitive effort rather than scoring, the study can more clearly isolate how the introduction of emojis shifts the processing speed and certainty of the recipient.

V. DISCUSSION

A. Theoretical Implications

The results of this study provide clear support for the original predictions derived from dual coding theory and visual salience research. The accuracy costs, reaction time increases, and confidence decreases all point in the same direction, that incongruent emoji and text pairings impose substantial cognitive load during emotional judgment. The pattern is consistent with a model in which two competing representations enter conflict and require an arbitration process that consumes additional resources before resolution is achieved. The size of the accuracy drop suggests that the influence of emoji on emotional judgment is not a marginal effect. Emojis also appear to interact with the processing of textual content in a way that is particularly disruptive when their valence is opposed, which is consistent with an interference account rather than a purely additive one. This was not explicitly predicted by either dual coding theory or visual salience research in their original formulations, and it represents a contribution that warrants further theoretical development.

The 31.4% error rate in the contradictory condition, and the fact that 95% of those errors reflected the emoji's valence rather than the text's, is the most important finding of the study. It provides direct support for the claim that visual cues dominate verbal cues during conflict resolution, and it extends Theeuwes's [2] account of salience into the domain of emotional and potentially verbal inference. The implication is that the distinctiveness of an emoji is sufficient to systematically bias emotional judgments, independent of the content of the text. This is a strong claim, and it should be treated with caution given the limitations of the present study, but it is a key of the observed error distribution. The individual question level data further supports this interpretation as items where the emoji's signal was particularly strong and the textual valence was ambiguous produced error rates as high as 66%, while items where the textual meaning was highly unambiguous produced zero errors. This suggests that the degree of visual dominance is determined and calibrated by the strength of the two competing signals.

The discrepancy between the control reaction time ($M = 9.15s$) and the congruent reaction time ($M = 7.15s$) also provides compelling evidence for the efficiency of multimodal processing. Even though the control was presumably simple to evaluate, the addition of a congruent emoji facilitated a 2.0 second reduction in response times on average. This suggests that emojis serve as accelerators that enable rapid categorization, whereas text only stimuli require a more sequential processing of logogens to reach those same emotional conclusion.

B. Implications for Communication Misunderstandings

For everyday communication, the findings carry a significant message. The emotional intent of a text message can be substantially overridden by a single emoji, even when the textual content is clear. Senders who deploy emojis ironically or sarcastically [6] may find that their intended meaning is

lost on a significant amount of recipients. This is not because those recipients lack the competence to understand irony or sarcasm, but because the visual signal captures their attention and biases their judgment before the textual meaning can assert itself. The individual question level findings underscore this point as even messages with relatively clear textual valence were misread by nearly half of participants when paired with a conflicting emoji. This suggests that the overriding effect is robust and not limited to genuinely ambiguous messages. The conditions under which recipients successfully resist this bias and the individual and contextual factors that support or undermine that resistance are important questions for future research.

C. Implications for Sentiment Analysis

The results argue against against a common assumption in sentiment analysis, which is that text and emoji can be treated as independent features whose contributions to message sentiment are additive [8]. The findings at hand however suggest that the relationship is more complex. When text and emoji agree, sentiment may be reliably inferred from either channel independently. But when they conflict, the emoji's influence is not additive; it is potentially dominant and overriding. A sentiment analysis model that weights text and emoji equally will systematically misclassify messages in which the two channels conflict, and in precisely the direction that the emoji's valence leans towards.

A more grounded approach might treat emoji valence as a paralinguistic cue [7] whose influence would appropriately scale with its cognitive salience, as this would explicitly model the conflict between channels and address the current gap in understanding communication dynamics [7]. The individual question variance observed in this study along with error rates ranging from 0% to 66% across stimuli, suggests that such a model would need to account for the specific emoji type, and the clarity of the textual signal, rather than treating all emoji and text conflicts as equivalent.

D. Communicative Norms

The nearly zero correlation between emoji use frequency and accuracy in the contradictory condition ($r = 0.004$) suggests that the visual override effect is not a habit, but likely an inherent property of the integration process. One interpretation is that the stimulus of the emoji is so potent that it can bypass any developed communicative norms. This implies that even those who encounter emojis daily are just as susceptible to the visual-verbal conflict as those who use them rarely.

The performance of older participants further supports this, as their high accuracy (80%–90%) despite varying familiarity levels suggests that the ability to resolve emoji and text conflict may be more closely tied to general attentional control than to specific digital communication habits.

VI. CONCLUSION

This study set out to investigate whether and how adults resolve emotional conflict when text and emoji convey op-

posing sentiments, and to characterize the cognitive mechanisms underlying that resolution. The results were again as follows. Contradictory emoji-text pairings imposed substantial cognitive costs across all three dependent variables. Classification accuracy dropped by 26 percentage points from the congruent to the contradictory condition. Reaction times in the contradictory condition were 2.32 seconds longer than in the congruent condition and 0.67 seconds longer than in the control condition. Confidence ratings fell from $M = 4.61$ in the congruent condition to $M = 4.00$ in the contradictory condition, indicating that participants were aware of their uncertainty rather than failing silently. Across 500 contradictory trials, 157 responses were misclassified, and in nearly every case the error reflected the emoji's valence rather than the text's, which is a pattern that points directly to visual dominance during conflict resolution.

Reflecting on this project, the most surprising finding was not that conflict imposed costs, but rather the degree to which individual participants varied in their ability to resist the visual signal. Error rates in the contradictory condition ranged from 0% to 60% across participants, which is a spread that suggests conflict resolution is not fixed but highly variable one. The individual question data adds another layer to this picture: error rates across contradictory stimuli suggested that the emoji type and the clarity of the textual signal interacted in meaningful ways that a single accuracy figure cannot capture. The inclusion of participants over the age of 24 suggests that this variability may have an age dimension as well, with emoji familiarity and communicative norms potentially playing a role in how readily the visual signal is suppressed. A followup study that manipulates participants' exposure to ironic texts and emoji use before the task, or that recruits a wider age range, could shed more light on these mechanisms than the measures used here.

More broadly, this study also demonstrates that emojis are not merely supplementary elements inside digital communication. They are cognitively potent signals that can override verbal content and introduce ambiguity into interactions that would otherwise be unambiguous. As AI systems increasingly attempt to infer emotional meaning from digital communication, understanding how humans resolve emoji and text conflict will remain a key research question. One that this study takes a small but concrete step toward answering.

VII. LIMITATIONS

The participant pool was relatively small and skewed toward college aged individuals, with approximately 80% of participants between 18 and 24 years old and the remaining 20% over the age of 24. While the inclusion of older participants adds a comparative dimension, the sample is insufficient to draw firm conclusions about age as a variable.

The experimental stimuli were deliberately stripped of conversational context. In real world communication, messages are embedded in conversational history, and context that provides cues not available to participants in the study. A message that produces conflict in isolation may be entirely

clear to a recipient who knows the sender's texting style and can further evaluate the emoji choice within the history of their relationship. The study here cannot speak to how these contextual factors affect emoji and text conflict resolution, and future research should examine this through field studies conducted within existing social relationships.

The stimuli used in the study drew exclusively from facial expression emojis with clearly valenced and widely agreeable emotional meanings. The broader emoji list includes gesture emojis, object emojis, animal emojis, and other symbols whose emotional valences are considerably more ambiguous. Whether the visual dominance effect generalizes to these less clear emoji types is an important question. It is plausible that the degree of visual dominance is itself calibrated by the clarity and salience of the emoji's emotional signal, such that highly expressive face emojis produce stronger override effects than other more ambiguous symbols.

Finally, while self reported emoji use frequency served as the primary measure of individual differences in visual cue reliance, it is an imprecise method for the underlying mechanisms. A more objective measure would provide stronger estimates of individual difference variables and enable more precise testing of the effects hypothesized within this paper.

Furthermore while a clear trend was observed, the reaction time data did not reach the threshold for statistical significance ($p=0.095$). Future research with a larger, more diverse sample size may be required to fully resolve the ins and outs of multimodal latency in digital communication."

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GITHUB LINKS

The custom web application used for survey collection and the analysis scripts used for the results section in this study are available at the following Georgia Tech GitHub repository: <https://github.gatech.edu/rbundy3/CognitiveScienceProject>

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Fig. 4. CITI Certificate